8.4 Technology in Space (SPC)

This unit focuses on a satellite whose remote sensing and other instruments are run from a solar power supply:

- illuminating solar cells
- operation of solar cells
- combining sources of emf
- radar imaging.

Mathematical models are developed to describe ohmic behaviour and the variation of resistance with temperature. Simple conceptual models are used for the flow of charge in a circuit, for the operation of a photocell, and for the variation of resistance with temperature.

Waves and photons are used to model the behaviour of light. Through a historical exploration of the photoelectic effect, students learn something of the provisional nature of scientific knowledge.

There are opportunities to develop ICT skills using the internet, spreadsheets and software for data analysis and display.

Through discussing the funding and execution of space missions, students have an opportunity to consider ethical and environmental issues and some of the decisions made by society regarding the use of technology.

Students will be assessed on their ability to:	Suggested experiments
29 identify the different regions of the electromagnetic spectrum and describe some of their applications	
69 define and use radiation flux as power per unit area	
67 recognise and use the expression $E = hf$ to calculate the highest frequency of radiation that could be emitted in a transition across a known energy band gap or between known energy levels	

Students will be assessed on their ability to:	Suggested experiments
66 use the non-SI unit, the electronvolt (eV) to express small energies	
64 recall that the absorption of a photon can result in the emission of a photoelectron	Demonstration of discharge of a zinc plate by ultra violet light
65 understand and use the terms threshold frequency and work function and recognise and use the expression $hf = \phi + ½ mv^2_{max}$	
63 explain how the behaviour of light can be described in terms of waves and photons	
71 explain how wave and photon models have contributed to the understanding of the nature of light	
50 describe electric current as the rate of flow of charged particles and use the expression $I = \Delta Q / \Delta t$	
51 use the expression $V = W/Q$	
52 recognise, investigate and use the relationships between current, voltage and resistance, for series and parallel circuits, and know that these relationships are a consequence of the conservation of charge and energy	Measure current and voltage in series and parallel circuits Use ohmmeter to measure total resistance of series/parallel circuits
53 investigate and use the expressions $P = VI$, $W = VIt$. Recognise and use related expressions, for example, $P = I^2R$ and $P = V^2/R$	Measure the efficiency of an electric motor (see also outcome 17)
54 use the fact that resistance is defined by $R = V/I$ and that Ohm's Law is a special case when $I \propto V$	

Students will be assessed on their ability to:	Suggested experiments
55 demonstrate an understanding of how ICT may be used to obtain current-potential difference graphs, including non-ohmic materials and compare this with traditional techniques in terms of reliability and validity of data	
56 interpret current-potential difference graphs, including non-ohmic materials	Investigate I-V graphs for filament lamp, diode & thermistor
70 recognise and use the expression efficiency = [useful energy (or power) output]/[total energy (or power) input]	
59 define and use the concepts of emf and internal resistance and distinguish between emf and terminal potential difference	Measure the emf and internal resistance of a cell, for example, a solar cell
60 investigate and recall that the resistance of metallic conductors increases with increasing temperature and that the resistance of negative temperature coefficient thermistors decreases with increasing temperature	Use of ohmmeter and temperature sensor
61 use $I = nqvA$ to explain the large range of resistivities of different materials	Demonstration of slow speed of ion movement during current flow
62 explain, qualitatively, how changes of resistance with temperature may be modelled in terms of lattice vibrations and number of conduction electrons	
46 explore and explain how a pulse-echo technique can provide details of the position and/or speed of an object and describe applications that use this technique	

Students will be assessed on their ability to:	Suggested experiments
47 explain qualitatively how the movement of a source of sound or light relative to an observer/detector gives rise to a shift in frequency (Doppler effect) and explore applications that use this effect	Demonstration using a ripple tank or computer simulation
48 explain how the amount of detail in a scan may be limited by the wavelength of the radiation or by the duration of pulses	
49 discuss the social and ethical issues that need to be considered, for example, when developing and trialling new medical techniques on patients or when funding a space mission	
72 explore how science is used by society to make decisions, for example, the viability of solar cells as a replacement for other energy sources, the uses of remote sensing	